2025 IEEE International Conference on Emerging Technologies and Applications (MPSec ICETA) | 979-8-3315-2131-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/MPSecICETA64837.2025.11118700

Design and Development of Machine Vision-Driven Robotics System for Safe and Efficient Inner Pipe Inspection.

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***Abstract - This* study focuses on developing an inspection robot, integrated with machine vision technology to identify and detect abnormalities on the inner surface of industrial pipelines. This robot can overcome the limitations faced by traditional methods. This robot is powered by Direct current motors on each wheel. To equip the robot with image capturing and object detection capabilities, it was integrated with a machine learning model trained in a Tensor-Flow environment using Python programming. Traditional methods required human presence on site with the implementation of this technology, humans can remotely control the robots functioning via a Bluetooth module and the captured data is displayed on a real-time data monitoring system. With capabilities such as autonomous navigation and advanced image processing techniques, this research aims to develop a robot to enhance safety, efficiency, reduced labor cost, and accessibility of pipe inspection robots for industrial applications. A prototype was developed to demonstrate the stated benefits and tests were performed in a practical environment to ensure robustness and reliability of the system for numerous industrial environments.**

Keywords— Pipe inspection, Machine vision, Anomaly Detection, Machine Learning, Real-time inspection component

I. Introduction

Inspection of industrial equipment at regular intervals is essential to maintain smooth and safe operations with pipeline inspection being a critical part of most industries. Due to the working environment of industries, industrial pipelines are prone to damage such as holes or cracks and blockages. These factors can pose significant risks to an industrial environment, and humans and can result in the entire industry being compromised [1]. Traditional methods involved manual inspection which was less safe and labour-intensive. The working environments include low oxygen levels which can affect human health [2]. Another disadvantage of manual inspection includes complex pipeline structures which make certain regions inaccessible and result in inefficient inspections [4].

Over the past few decades, technological advancement has given rise to robot technology which has majorly been employed for industrial use, replacing traditional methods. For pipeline inspection, robots are integrated with vision cameras and navigational abilities that investigate the pipelines providing a safer and efficient alternative as compared to manual inspections [3]. The focus of this study is to design and fabricate a pipeline inspection robot equipped with machine vision technology and navigational capabilities to identify and detect different types of defects present in a pipe.

The proposed inspection robot is designed in SolidWorks and trained in Python programming [5]. A Raspberry Pi camera is mounted on the robot to capture and detect and a Bluetooth module is placed to allow for remote operation [6]. Captured data is displayed on a screen allowing the operator to observe the live data, enhancing their ability to make informed decisions [8]. The goal of this study is to develop a pipeline inspection robot with machine vision and navigational capabilities that can identify numerous pipe abnormalities. The captured high-resolution images are analysed by the machine vision algorithm and are compared to the pre-existing database.

II. Machine Vision

A. Introduction

Machine vision is a technology that combines hardware and software components to execute numerous automation tasks related to image capturing and processing. This technology integrates AI (Artificial Intelligence), numerous programming methods and algorithms [7]. It utilizes digital sensor fuses that are embedded in industrial cameras to capture images that are further processed and analysed to make informed decisions regarding the region of interest. Machine vision technology is much preferred in the electrical and electronic industry to perform inspections for PCBs (Printed Circuit Boards), solar wafers, and wire bonding for ICs (integrated Circuits) due to its exceptional operational speed, accuracy and improved efficiency [9]. The components of a machine vision system are a Vision processing unit, Image Sensors, Lenses, Lighting equipment and communication devices [10].

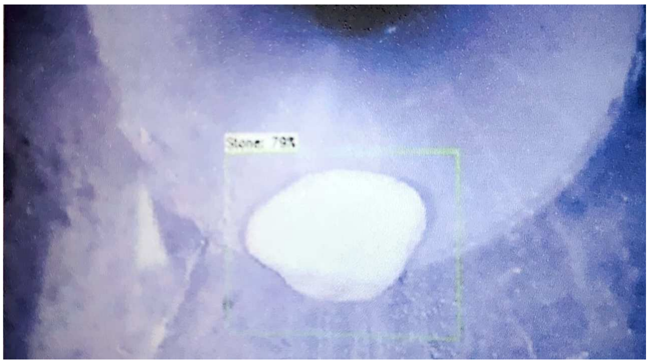

Fig. 1. Surface irregularities identified by the Inspection robot are highlighted using the green box.

Lighting equipment is used to provide sufficient light that can be illuminated to allow the lens to capture images of interest. These captured images are converted to digital signals via the image sensor and sent to the vision processing unit where data filtration and region of interest are extracted to make informed decisions and communicate this information via I/O (Input-Output) signals [11].

Machine vision performs a few operations to capture an image and to perform pattern recognition as provided in Figure 2. The first step involves ADC (Analogue to Digital Conversion) of the captured image followed by filtration and contrast enhancement by a process known as image processing [12]. The third step includes extraction of the region of interest followed by image analysis considering measurement and relationships of objects. The final step involves comparing the region of interest with a pre-existing data set by a process known as pattern recognition.

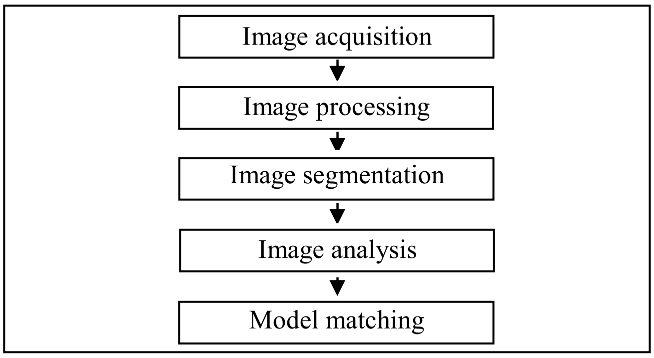

Fig. 2. Machine vision process flow (Various steps performed from image acquisition to irregularity detection and matching).

B. Machine Vision Python Program

The program is developed in a TensorFlow environment using a Python Programming application [13]. The program aims to do a live inspection of the pipe from the inside and identify pipe blockage or defects such as cracks, holes or stones. The identified object is displayed by a green box on the live screen as displayed in Figures 1, 4 and 5. The developed program is capable of showcasing the similarity index on top of the identified region indicating the percentage of data matching the live assessment data.

III. Image Processing

A. Introduction

Image processing involves a process of processing captured images, extracting valuable information and enhancing these images using a digital computer. It involves image acquisition, image restoration, image enhancement, compression, segmentation, image processing, image restoration, image colour processing and so on [14]. Image processing includes the following steps:

1. Using image acquisition tools to import images.
2. Analysing and manipulating the imported image.
3. Producing output images based on analysis.

B. Image Processing Flowchart

Figure 3 displays the flow of the image processing process indicating various steps and conditional blocks.

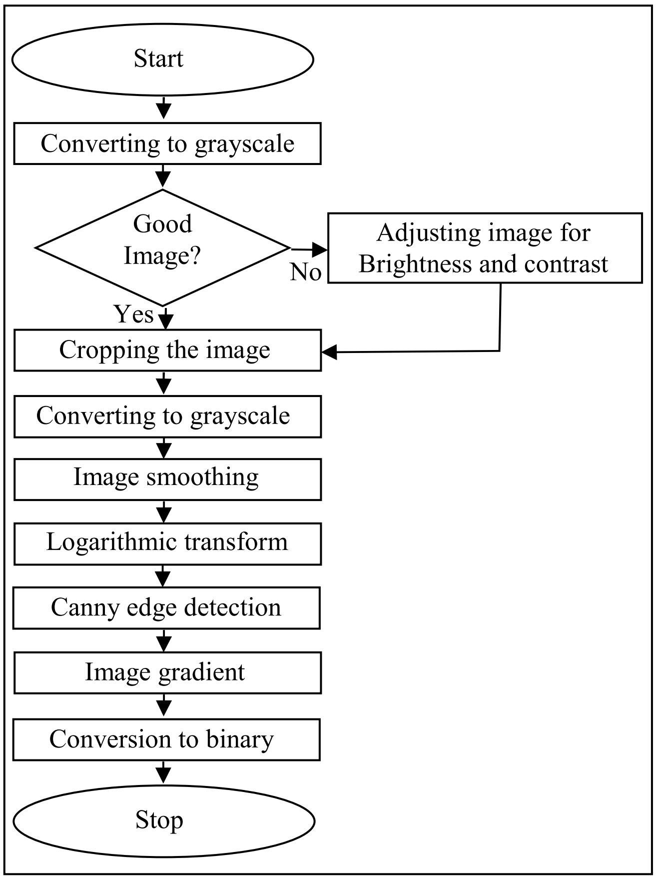

Fig 3. Flow chart of Image Processing.

The first step of image processing is capturing images image capturing followed by cropping, converting to a grayscale image, image soothing, logarithmic transform, canny edge detection, image gradient, and lastly converting to a binary image (or binary data).

Once the image is converted to a suitable standard, it is cropped to cut unnecessary image portions. This aids in directing the focus only on areas of interest for further analysis.

The next step is to convert the image to grayscale, known as thresholding. It is the process of converting only the portion or the entire pixel values to zeros or the maximum value i.e., 255. Pixels below threshold are converted to zeros and those above thresholds are converted to 255. This can be better understood as all the pixel values having zero value refer to black and all the pixels converted to 255 refer to white.

Following the grayscale procedure, the next step is image smoothening. Image smoothening involves noise filtration and conversion to blur images which can significantly aid in pre-processing and post-processing of deep learning models. Logarithmic transform or applying gradient involves changing the rate of pixels in the X or Y direction or in both directions to detect edges and identify sudden changes in the pixel values.

Canny edge detection performs Gaussian noise filtering, vertical and horizontal edge detection, removing unwanted edge points and hysteresis thresholding. Changing the minimum and maximum values of the canny image yields better results. Image contouring contains images that have continuous outlines which are identified by using the 'OpenCV' function and its coordinates are collected. These are essential for detecting object shapes, motion detection, and image segmentation.

The binary image is an image that has only two pixels either "0" or "1" i.e., either black colour or white colour. It is also called a bi-level or two-level. The binary image allows distance transformation plays an important role in image recognition and orientation-free matrixes. All the image processing transformations mentioned above can be carried out in any sequence. The outputs generated by the above transformations are independent and can be used to extract various forms of data for image enhancement.

The program plots graphs of the number of pixels vs grey levels for both input and output images. Input images usually have only two values in the graph representing the maximum and minimum values of the pixels i.e., 0 and 255. The output image usually has values between 0 and 255 representing all colours such as red, green and blue or a combination of all three (RGB).

C. Image Processing Python Program

Similar to a Machine vision program, an image processing program is developed in a tensor flow environment using Python programming for detecting numerous abnormalities from the inner surface of the pipe under inspection. This program detects and captures abnormalities like holes, cracks, corrosive surfaces and blockage in pipes. Upon identification, the program enables the robot to capture these images as blur, grayscale image, binary image, logarithmic transformation or contour image. The output produced is a digital image as displayed in the below figures.

D. Image Processing Python Program

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Fig. 4. Crack identification by Inspection Robot highlighted using the green box.

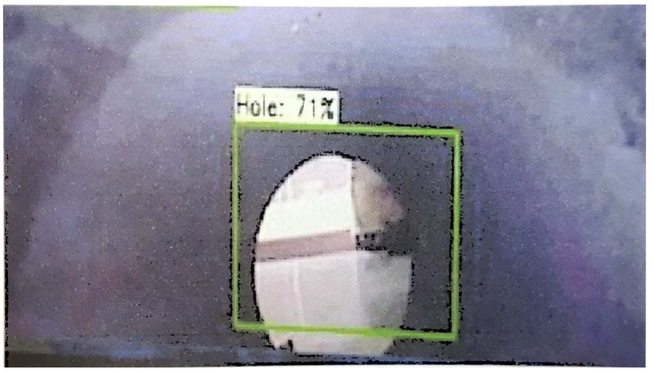

Fig. 5. Hole identification by Inspection Robot highlighted using the green box.

IV. COMPONENTS

Numerous components were used for the successful completion of the project. The list of components along with their specifications is discussed below:

A. Raspberry Pi 3

The Raspberry Pi 3 has four built-in USB (Universal Serial Bus) ports to provide enough connectivity for a mouse, keyboard, and so on. Powering the Raspberry Pi 3 is easy, just plug any USB power supply into the micro-USB port. There's no power button so the Pi will begin to boot as soon as power is applied, to turn it off simply remove power. The four built-in USB ports can even output up to 1.2A enabling to connect more power-hungry USB devices. On top of all that, the low-level peripherals on the Pi make it great for hardware hacking. The 0.1" spaced 40-pin GPIO header on the Pi gives you access to 27 GPIO (General Purpose Input-Output), UART (Universal asynchronous receiver/transmitter), I2C (Inter-Integrated Circuit), SPI (Serial Peripheral Interface) as well as 3.3 and 5V sources. Each pin on the GPIO header is identical to its predecessor the Model B+. The biggest change that has been enacted with the Raspberry Pi 3 is an upgrade to the next-generation main processor and improved connectivity with Bluetooth Low Energy (BLE) and BCM43143 Wi-Fi on board.

Additionally, the Raspberry Pi 3 has improved power management, with an upgraded switched power source of up to 2.5 Amps, to support more powerful external USB devices. The main differences are the quad-core 64-bit CPU onboard Wi-Fi and Bluetooth. The RAM (Random Access Memory) remains at 1GB and there is no change to the USB or Ethernet ports. However, the upgraded power management should mean the Pi 3 can make use of more power-hungry USB devices.

B. Raspberry Pi Camera

This is a high-definition camera module providing high sensitivity, low crosstalk and reduced image noise in an ultra-small and lightweight design. This camera is compatible with most of Raspberry Pi models. This module connects to Raspberry Pi via a CSI (Camera Serial Interface) connector designed specifically for interfacing with the camera. The CSI bus is capable of extremely high data rates and it communicates pixel data to the processor.

C. Channel Relay

This component is a low-level 5V 4-channel relay interface board with each channel requiring 1520mA of DC and is equipped with high-current relays that operate under AC250V 10A or DC30V 10A. This can be used to control various appliances and equipment. Channel relays have a standard interface controlled via a microcontroller and are optically isolated from high voltage as a part of safety requirements and to prevent ground loops. The relay boards with 3.3V signals can be utilized if JD-VCC (Relay Power) is provided with a 5V supply and the VCC (Voltage Common Collector) to the JD-VCC jumper is eliminated. This 5V relay supply can be isolated from a 3.3V device or can be provided with a common ground if opto-isolation is neglected.

D. DC Johnson L298 Motor

This is a high-torque motor commonly used in Pan/Tilt cameras, auto shutters, welding machines, water meter IC (internal circuit) cards, grills, ovens, cleaning machines garbage disposers, household appliances, slot machines, money detectors, automated actuators.

E. LED

The LED (light-emitting diode) is a widely used lighting system used for various applications due to its low power consumption ability, longer life and low cost.

F. Battery

The majority of motors utilize either a 12 or 24-volt power source. Hence to ensure a longer life of the motor for the required application, the battery and motor need to be sized correctly. Lead acid batteries are commonly used due to their low cost and availability.

G. Bluetooth Module

Bluetooth module is a device that establishes a wireless connection between the robot and the remote device.

H. Wheels

Wheels provide support to the robot and aid in locomotion. They provide the required rotational motion to move the robot in forward and reverse direction. They are made up of plastic and coated with rubber to avoid slippage and dampen vibrations.

V. CALCULATIONS

Selection of a suitable motor is critical for this project as the motor must be able to overcome the entire robot load and external forces acting on the wheels and robot. Below are the calculations performed for the selection of a motor. This selection is considered from the design point of view and the weight of the robot is considered as 10 Kg.

A. Voltage Requirement

A specified amount of voltage is to be supplied to the motor that provides rotational motion to move the robot assembly. The voltage is calculated as follows:

$$V = I \times R \qquad (1)$$

Where I = Supplied Current (in Ampere)
R = Resistance (in ohm)

Assuming required voltage = 12V and current = 7.5 amp

$$R = V/I = 12/7.5 = 1.66 \text{ ohm}$$

B. Torque Requirement

The motor needs to exceed a certain amount of load to provide rotational movement to the robot. This can be achieved if the motor is capable of producing a torque that can match the requirement to provide motion to the wheels. The torque required is evaluated as follows:

$$T = f \times r \qquad (2)$$

Where, f = Force due to gravity and robot weight (in N)
r = Radius of wheel (in m)
T = Torque (in Nm)

The radius of a selected wheel, r = 0.05 m
Hence from equation (1),

$$T = 10 \times 9.81 \times 0.05 = 4.905 \text{ Nm}$$

C. Power Requirement

Based on the evaluated torque, the output power required to overcome the torque is as follows:

$$P = \frac{2\,\pi\,N\,T}{60} \qquad (3)$$

Where, T = Torque
N = Speed of motor (in rpm)
P = Power (in Watts)

Alternatively, power can be evaluated using the formula below.

$$P = V^2 / R \qquad (4)$$

Hence to find the power requirement, equation (4) is evaluated as follows:

$$P = (12)^2/1.66 = 86.94 \text{ Watts} \approx 90 \text{ Watt}$$

D. Speed (rpm) requirement

After calculating the power requirement, the motor speed achieved is evaluated using equation (3), hence.

$$90 = \frac{2 \times \pi \times N \times 4.905}{60}$$

Hence, N = 175.4 rpm
The above speed value is for forward and backward movement.

E. Heat Dissipation

Motor during its operation increases internal temperature and dissipates some energy in the form of heat. This heat is generated due to total resistance which is induced due to friction losses in the stator and rotor. Hence, Power dissipation (P_{dis})can be evaluated as follows:

$$P_{dis} = I^2 \times R \qquad (5)$$

$$P_{dis} = (7.5)^2 \times 1.66 = 93.375 \text{ w}$$

F. Spring

A suitable market-available tension spring was acquired that can perform the function of extending the robot legs when it moves from a region of a smaller diameter to a larger diameter.

VI. Material Selection

As discussed in the previous sections, the final developed robot can be employed for various applications. This robot can be subjected to random or continuous internal pipe environment changes, body wear and tear, and possible breakdown. To ensure that the robot is able to withstand extreme conditions, FRP (Fiber-reinforced polymer) and fiberglass-reinforced polymer materials are used.

FRP was selected for its excellent strength-to-weight ratio capability, resistance to corrosion and external forces or impacts, durability, flexibility and versatility. These types of materials are non-magnetic and non-conductive in nature, offering superior thermal and electrical insulation. Most of the parts in this robot are 3-D (Three-dimensional) printed except for electrical and electronic components and the body as shown in Figure 9 and Table IX.

The body as shown in Figure 9, part code 'K', is the primary part that performs the function of housing and bearing the load of all other components. To ensure that the body is capable of performing its intended function, fiberglass-reinforced plastic is used due to its high corrosion resistance properties from harmful chemicals and unpredictable environmental change.

VII. Model Development

A three-dimensional design of the robot was developed in SolidWorks Software. The design procedure included concept development followed by Pugh's evaluation matrix which aided in selecting the most suitable and affordable configuration. The developed three-dimensional model is displayed in Figures 6, 7 and 8. Figure 9 and the table together provide data about individual parts along with the material used.

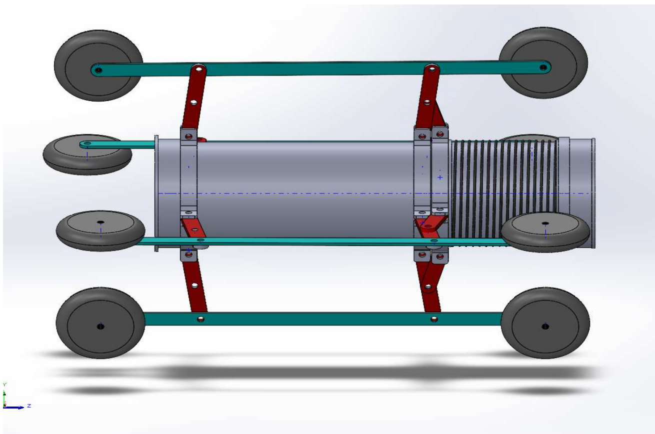

Fig. 6. Side of Pipe Inspection Robot (This design showcases components such as wheels (in grey), primary links (in red), secondary links (in blue), and spring elements (on the right appears as a threaded element).

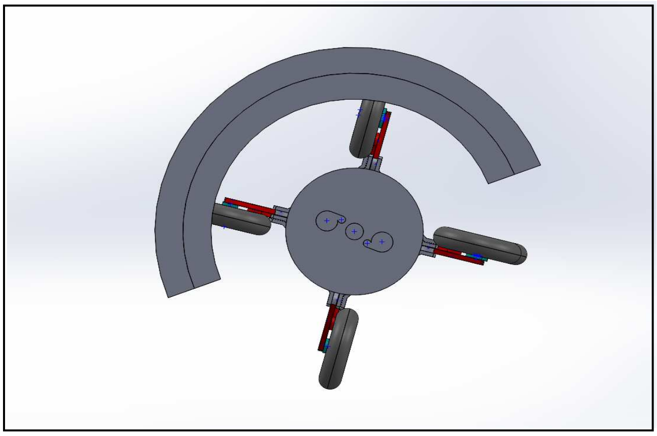

Fig. 7. Front view of Inspection pipe robot inside a test pipe (The test pipe is represented as an arc is a varying diameter with the inspection robot navigating through it).

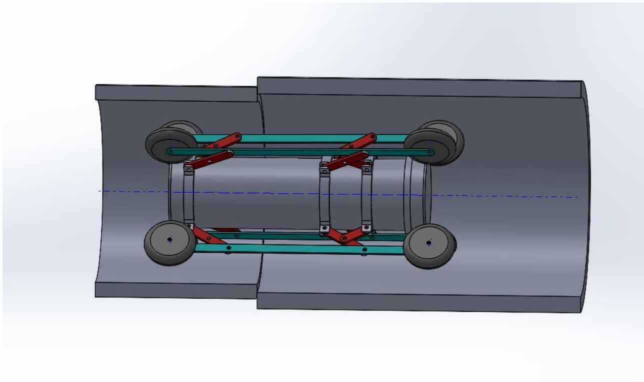

Fig. 8. Front view of Inspection pipe robot inside a test pipe (the robot can be observed navigating through varying pipe diameters and the spring element is in action and compresses as pipe diameter decreases).

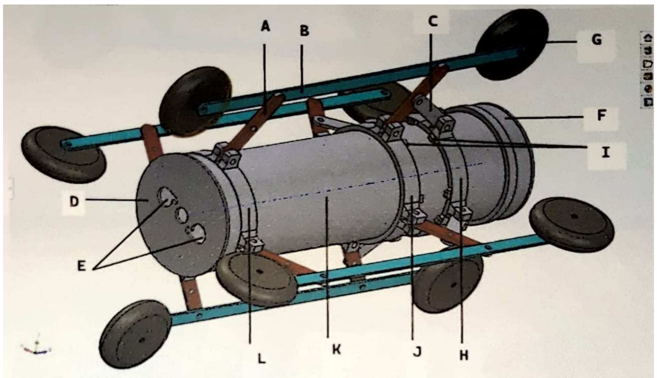

Fig. 9. Pipe Inspection Robot Nomenclature.

TABLE I. Pipe Inspection Robot Nomenclature

Part Code	Part Name	Material
A	Link 1	Fiberglass Reinforced Plastic
B	Link 2	Fiberglass Reinforced Plastic
C	Link 3	Fiberglass Reinforced Plastic
D	Front Panel	Fiber Reinforced Polymer
E	Camera Mount	Fiber Reinforced Polymer
F	Access Panel	Fiber Reinforced Polymer
G	Wheel	Rubber
H	Transitional element	Fiber Reinforced Polymer
I	Operating envelope	Steel Spring
J	Limit Clamp 1	Fiber Reinforced Polymer
K	Body	Fiberglass Reinforced Plastic
L	Limit clamp 2	Fiber Reinforced Polymer

VIII. Results and Discussions

The primary objective of this project was to develop a robot that can navigate through industrial pipelines to identify different types of defects like cracks, holes, blockages, etc., and compare the captured images with pre-loaded data. The following are the results of the experiment:

The robot was equipped with an 8 Megapixel Raspberry Pi Camera which successfully captures surface defects in the resolution of 3280x2464 pixels with a frame rate of 30 fps as shown in Figures 1, 4 and 5. Captured images were sufficiently clear under varying lighting conditions demonstrating the robustness of the integrated LED system.

The developed robot showcased the capabilities of adapting to varying pipe diameters in the range of 120 mm to 400 mm as displayed in Figure 7. The spring mechanism ensured effective gripping by pushing the wheels towards the inner surface of the inspected pipe. The robot was capable of travelling in horizontal, vertical and diagonal directions.

The image-matching algorithm maintained an accuracy rate of 80% and above in identifying cracks, holes and blockages from the database. The algorithm performed image processing with an average of 0.8 seconds per image as displayed in figure 10, 11.

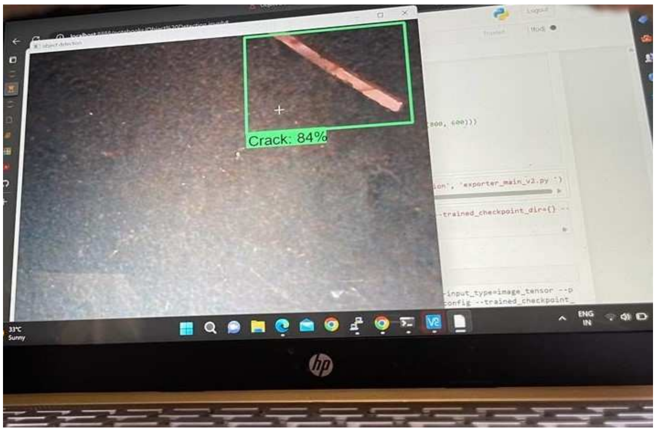

Fig. 10. Crack Identification (Crack identification by inspection robot and comparing the captured image with the database resulting in 84% accuracy in identifying fault in the pipe).

Fig. 11. Blockage (Stone) Identification (Blockage identification by inspection robot and comparing the captured image with the database resulting in 99% accuracy in identifying fault in the pipe).

IX. Conclusion

This research project demonstrates the potential of robotic technology to enhance industrial pipe inspection efficiency and safety. This developed technology addresses manual inspection limitations and provides advanced capabilities through machine vision and autonomous operation. This innovative approach offers significant benefits to industries including reduced labor costs, improved safety, and enhanced inspection accuracy. As the project progresses, further refinement and field tests are conducted to ensure the robustness and reliability of the system in various industrial environments.

X. Acknowledgment

We would like to thank our parents for their contribution and support throughout the project phase. We would also like to express our gratitude to Mr. Devashish Colvalkar, Mr. Sarvesh Sinai, Mrs. Tarni Desai and Prof. Laxmikant Bordekar for their contribution and constant help throughout the project.

XI. References

[1] Singh, A. V., Gupta, V. K., & Mukherjee, S. (2018). Development of in-pipe inspection robot. Materials Today: Proceedings.

[2] Bogue, R. (2019). Applications of robotics in test and inspection. Industrial Robot: An International Journal, 45.

[3] Addams, W., Sridhar, S., Talman, C. M., & Elsad, H. (2018). Water pipe robot using soft inflatable actuators. In 2018 IEEE International Conference on Robotics.

[4] Ab Rashid, M. Z., Salim, S. A. Z. S., Mamat, N., Munawwarah, S., & Putra, S. M. (2020). Modelling of the in-pipe inspection robot: A comprehensive review. Ocean Engineering, 203, 107206.

[5] Zhang, Y., & Yan, G. (2007). In-pipe inspection robot with active pipe diameter adaptability and automatic tractive force adjusting. Mechanism and Machine Theory, 42(12), 1618-1631.

[6] Ciszewski, M., & Buratowski, T. (2018). Modelling, simulation and control of a pipe inspection mobile robot with an active adaptation system. IFAC PapersOnLine, 51(22), 132-137.

[7] Liu, Z., & Krys, D. (2012). The use of a laser range finder on a robotic platform for pipe inspection. Mechanical Systems and Signal Processing.

[8] Kerakalamatti, M., Sudhir, R., & Kadam, P. (2017). Design and development of pipe climbing robot. International Journal of Magazine and Engineering, Technology, Management and Research, 4(3).

[9] Mishra, D., Agrawal, K. K., Abbasi, A., Srivastava, R., & Yadav, R. S. (2019). PIG [Pipe Inspection Gauge]: An artificial dustman for cross-country pipelines. Procedia Computer Science, 152, 333-340.

[10] Alhassana, A. B., Zhang, X., Shen, H., & Xu, H. (2020). Power transmission line inspection robots: A review, trends and challenges for future research. Electrical Power and Energy Systems.

[11] Roslin, N. S., Anuar, A., Abdul Jalal, M. F., & Mohamed Sahari, K. S. (2012). A review: Hybrid locomotion of in-pipe inspection robot. Engineering Procedia, 2012 International Symposium on Robotics and Intelligent Sensors.

[12] Yeh, C.-Y., Chen, C.-Y., & Juang, J.-Y. (2020). Soft hopping and crawling robot for in-pipe travelling. Extreme Mechanics Letters. https://doi.org/10.1016/j.eml.2020.100854

[13] Liu, T., Pei, C., Cai, R., Li, Y., & Chen, Z. (2020). A flexible and noncontact guided-wave transducer based on coils-only EMAT for pipe inspection. Sensors and Actuators A: Physical, 314, 112213.

[14] S. Khan, A. Ali, and T. Murata, "Development of autonomous pipe inspection robot with adaptive movement mechanism for pipeline maintenance," IEEE Access, vol. 8, pp. 148767–148776, 2020.